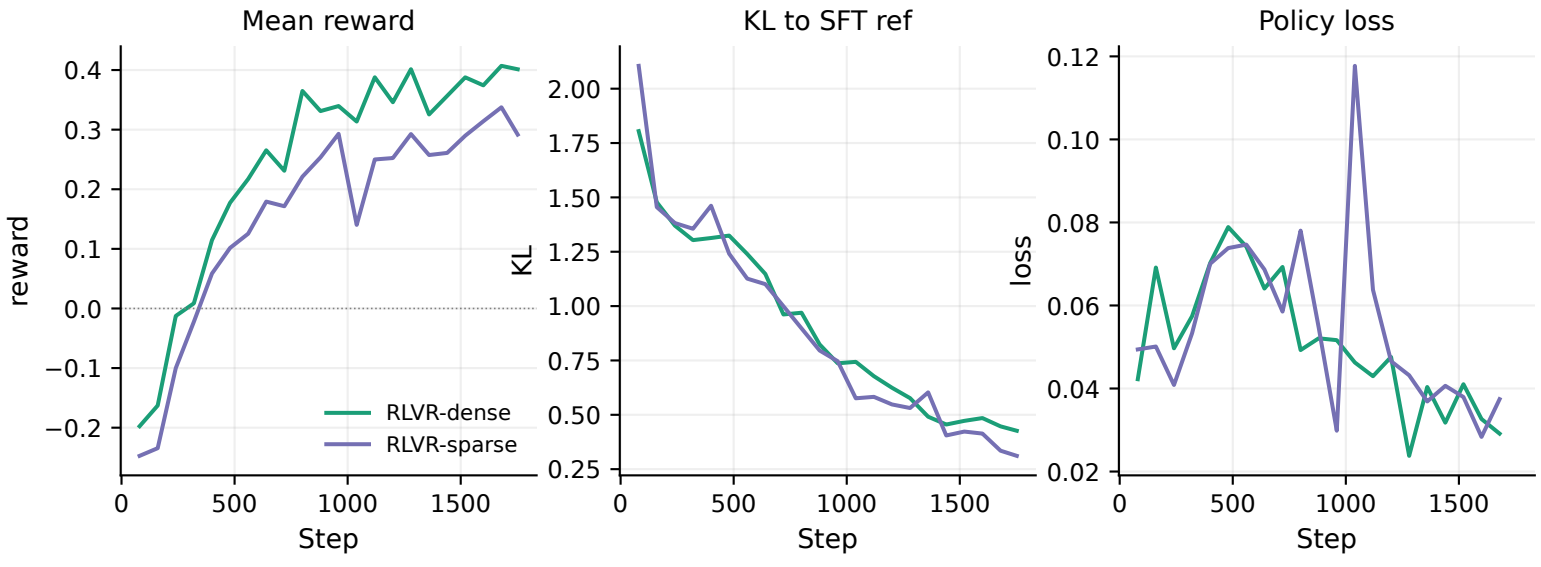
RLVR training stability — 4 epochs over 3506 prompts (G=4, T=1.0)
Mean reward
KL to SFT ref
Policy loss
reward
KL
loss
Step
Step
Step
RLVR-dense
RLVR-sparse
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.12
0.10
0.08
0.06
0.04
0.02
0
500
1000
1500